

FMUSER FUQ-200 Head-End Processor 3-Slot Modular 1 GE RJ45 Port MPEG-2-H.264-HEVC DVB-S2-T2-ISDB-T Multi-Module Broadcast Processing for Professional DTV Systems



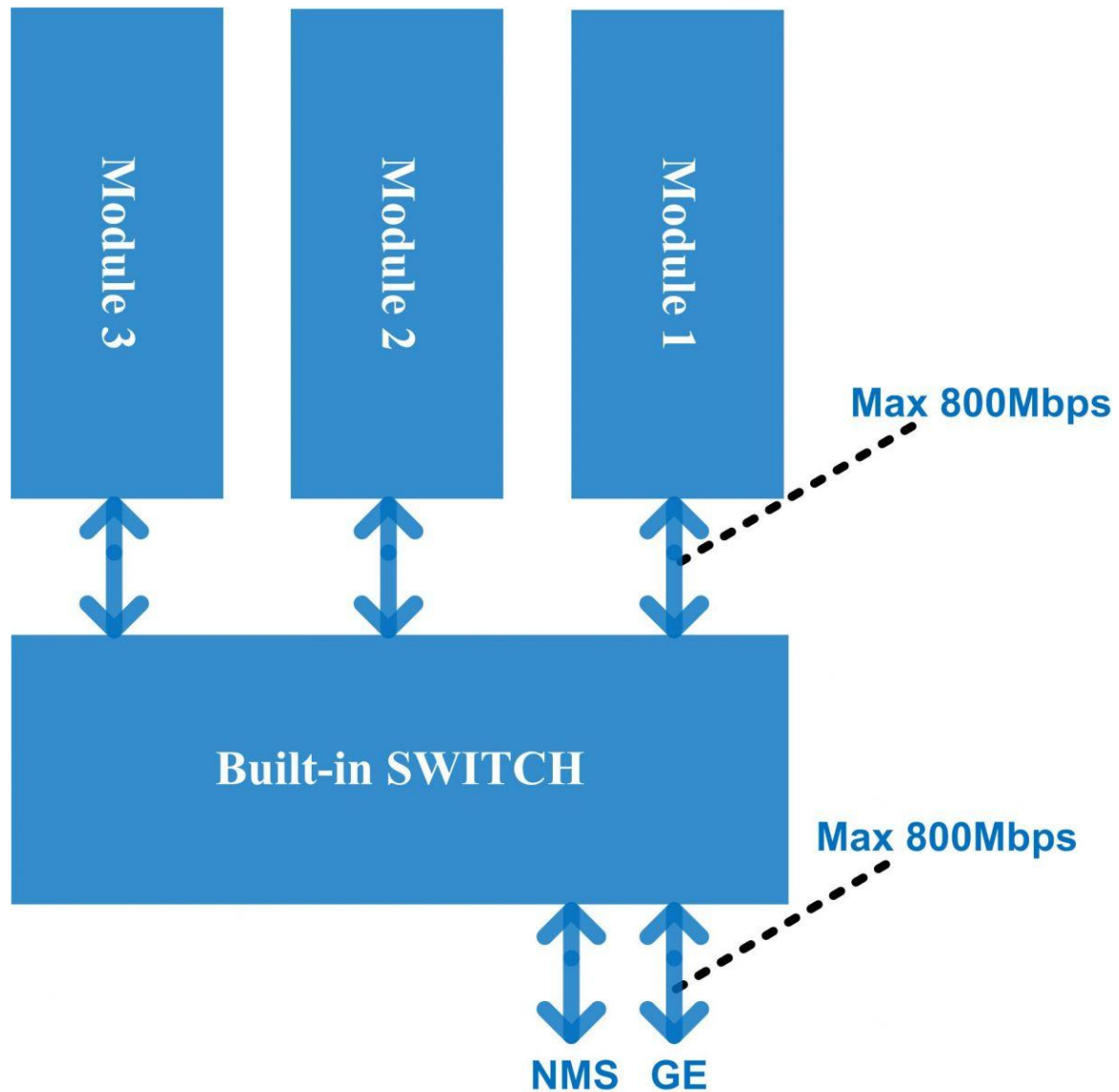
Product overview

The FMUSER FUQ-200 head-end processor handles encoding, decoding, transcoding, multiplexing, descrambling, and modulating via modular inputs. It processes broadcast signals through flexible module combinations. This unit features a 1U chassis with 3 slots and a 1 GE bi-directional data port (RJ45 interface). Ideal for professional DTV systems and broadcast center workflows.

Key feature

- Supports modular plug-in design with 1U chassis and 3 slots.
- Enables flexible combination of encoding, decoding, transcoding, multiplexing, descrambling, and modulating modules.
- Delivers high-capacity 800M data processing.
- Features 1 GE bi-directional data port (RJ45 interface) for robust IP connectivity.
- Provides Web-based management and firmware updates.

Principle Chart



Module Specifications:

4 CVBS/SDI Encoding Module (DX214B)

Module Specifications: Input: 4 CVBS (DB9 to RCA) or 4 SDI (BNC) Output: 1MPTS and 4 SPTS output over UDP/RTP, unicast and multicast Video Encoding: Video format: MPEG-2, MPEG4 AVC/H.264 Image format: PAL, NTSC SD signal (Only for CVBS input) Resolution: Input: 720*576 -50i Output: 720*576/352*288/320*240/320*180/176*144/160*120/160*90-50Hz Input: 720*480 -60i Output: 720*480/352*288/320*240/320*180/176*144/160*120/160*90-60Hz Rate Control: CBR/VBR GOP structure: IPPP, IBPBP, IBBPB, IBBBP Video bitrate: 0.5~5Mbps Audio Encoding: Audio format: MPEG1 Audio Layer 2, LC-AAC, HE-AAC Sampling rate: 48KHz Bit-rate: 48/56/64/80/96/112/128/160/192/224/256/320/384 Kbps Support Logo, Caption, QR Code insertion

4 CVBS Encoding Module (DX214/214A)

Module Specifications: Input: 4 CVBS video, 4 Stereo Audio (DB9 to RCA) Output: 1MPTS and 4 SPTS output over UDP/RTP, unicast and multicast Video Encoding: Video format: MPEG-2 (4:2:0) Image format: PAL, NTSC SD signal Output resolution: 720×480_60i, 544×480_60i, 352×480_60i, 352×240_60i, 320×240_60i, 176×240_60i, 176×120_60i, 720×576_50i, 704×576_50i, 640×576_50i, 352×288_50i, 320×288_50i, 176×288_50i, 176×144_50i GOP structure: IP, IBP, IBBP, IBBBP Video bitrate: 0.1Mbps~8Mbps per channel Support CC (closed caption) Audio Encoding: Audio format: MPEG-1 Layer 2, DD AC3 (2.0) Sampling rate: 48KHz Audio bitrate: 128/192/256/320/384kbps each channel Support Logo, Caption, QR Code insertion (for DX214A only)

8 CVBS Encoding Module(DX218S)

Module Specifications: Input: 8 CVBS video, 8 Stereo Audio (DB15 to RCA), or 8 ch analog audio input to work as audio encoder module Output: 1 MPTS and 8 SPTS output over UDP/RTP, unicast and multicast Video Encoding: Video format: MPEG4 AVC/H.264 Image format: PAL, NTSC SD signal Resolution: 720×576i, 720×480i Rate Control: CBR/VBR GOP structure: IPP Video bitrate: 1~7Mbps each channel Audio Encoding: Audio format: MPEG-1 Layer 2 Sampling rate: 48KHz Bit-rate: 64/128/192/224/256/320/384Kbps each channel Support Logo, Caption, QR Code insertion

2 HDMI/SDI Encoding/Transcoding Module (DX202A/DX202A-D)

Module Specifications: Input: 2*HDMI, 2*BNC for CC (Closed Caption) input - (DX202A) Input: 2*HD/SD SDI - (DX202A-D) Output: 1*MPTS output over UDP, Unicast/Multicast Video Encoding: Video format: MPEG2 & MPEG4 AVC/H.264 Input resolution: 1920*1080_60P, 1920*1080_50P, 1920*1080_60i, 1920*1080_50i, 1280*720_60p, 1280*720_50P, 720*480_60i, 720*576_50i Aspect ratio: 16:9, 4:3 Video bitrate: 0.8~19Mbps Support CC (closed caption) Audio Encoding: Audio format: MPEG1 Layer II, MPEG2-AAC, MPEG4-AAC, Dolby Digital AC3 (2.0) encoding (Optional); AC3 (2.0/5.1) passthrough Sampling rate: 48KHz Audio bitrate: 64Kbps-320kbps each channel Video Transcoding: 2*MPEG2 HD → 2*MPEG2/H.264 HD; 4*MPEG2 HD → 4*MPEG2/H.264 SD; 2*H.264 HD → 2*MPEG2/H.264 HD; 4*H.264 HD → 4*MPEG2/H.264 SD; 4*MPEG2 SD → 4*MPEG2/H.264 SD; 4*H.264 SD → 4*MPEG2/H.264 SD Audio Transcoding: MPEG-1 Layer 2, AC3 (Optional) and AAC any-to-any

4 HDMI/SDI Encoding Module (DX224H/HV)

Module Specifications: Input: 4×HDMI (1.4) /SDI input, HDCP 1.4 Output: 1 MPTS and 4 SPTS output over UDP/RTP/RTSP Video Encoding: Video format: HEVC/H.265 & MPEG 4 AVC/H.264 Resolution: Chroma: 4:2:0 Bit rate (each channel): 0.5Mbps~20Mbps (H.265); 4 Mbps~20Mbps (H.264) Rate Control: CBR/VBR GOP structure: IBBP, IPPP Audio Encoding: Audio format: MPEG-1 Layer 2, LC-AAC, HE-AAC, HE-AAC V2, AC3 Passthrough Bit-rate (each channel): 48Kbps~384Kbps (MPEG-1 Layer 2 & LC-AAC); 24 Kbps~128 Kbps (HE-AAC); 18 Kbps~56 Kbps (HE-AAC V2) Sampling rate: 48KHz Support Audio Gain adjust Support Logo, QR Code insertion--Optional as per order

4 HDMI/SDI Encoding Module (DX224V)

Module Specifications: Input: 4×SDI/HDMI (1.4) input, HDCP 1.4 Video Encoding: Video format: HEVC/H.265 & MPEG 4 AVC/H.264 Resolution: HDMI V1 & SDI HDMI V2: Chroma: 4:2:0 Bit rate: 0.5Mbps~20Mbps (H.265); 4 Mbps~20Mbps (H.264) Rate Control: CBR/VBR GOP structure: IBBP, IPPP Audio Encoding: Audio format: MPEG-1 Layer 2, LC-AAC, HE-AAC, HE-AAC V2, AC3 Passthrough Sampling rate: 48KHz Bit-rate (each channel): 48Kbps~384Kbps (MPEG-1 Layer 2 & LC-AAC) 24 Kbps~128 Kbps (HE-AAC), 18 Kbps~56 Kbps (HE-AAC V2) Support audio gain adjust Support Logo, QR Code insertion--available on HDMI V1

2 Tuner Descrambling Module (DX902A)

Stream in: 2 Tuner input, F Type, 16 IP inputs from data port, UDP/RTP, unicast and multicast, IGMP V2/V3 Stream out: 1 MPTS and 16 SPTS outputs over UDP/RTP/RTSP, Unicast/Multicast DVB-CI: 2 independent common interface slots Tuner input: DVB-S/S2/S2X Symbol rate: QPSK/8PSK/16APSK 0.5~45 Msps; 8APSK/32APSK: 0.5~40Msps Input Frequency: 950-2150MHz DVB-S: Constellation: QPSK FEC Demodulation: 1/2, 2/3, 3/4, 5/6, 7/8 DVB-S2: Constellation: QPSK/8PSK/16APSK/32APSK FEC Demodulation: QPSK: 1/2, 2/3, 3/4, 5/6, 3/5, 4/5, 8/9, 9/10 8PSK: 3/5, 2/3, 3/4, 5/6, 8/9, 9/10 16APSK: 2/3, 3/4, 4/5, 5/6, 8/9, 9/10 32APSK: 3/4, 4/5, 5/6, 8/9, 9/10 DVB-S2X: Constellation: QPSK/8PSK/8APSK/16APSK/32APSK FEC Demodulation: QPSK: 1/2, 3/5, 2/3, 3/4, 4/5, 5/6, 8/9, 9/10, 13/45, 9/20, 11/20 8PSK: 3/5, 2/3, 3/4, 5/6, 8/9, 9/10, 23/36, 25/36, 13/18 8APSK: 5/9-L, 26/45-L 16APSK: 2/3, 3/4, 4/5, 5/6, 8/9, 9/10, 1/2-L, 8/15-L, 5/9-L, 26/45, 3/5, 3/5-L, 28/45, 23/36, 2/3-L, 25/36, 13/18, 7/9, 77/90 32APSK: 3/4, 4/5, 5/6, 8/9, 2/3-L, 32/45, 11/15, 7/9, 9/10 Support Diseqc function Multiplexing: Maximum PID Remapping: 256 output pids Function: PID remapping (automatically or manually), Accurate PCR adjusting, generate PSI/ SI table automatically Descrambling: CAM/CI Quantity: 2 BISS Mode: Mode 1, Mode E; up to 120Mbps, 32 BISS key

4 FTA Tuner Module (DX904B)

Module Specifications: Stream in: 4 Tuner input, F Type Stream out: 1 MPTS out for mux, 4 MPTS out for 4 tuner input pass through over UDP/RTP/RTSP, unicast/multicast Tuner input: DVB-S/S2/S2X Symbol rate: QPSK/8PSK/16APSK 0.5~45 Msps; 8APSK/32APSK: 0.5~40Msps Input Frequency: 950-2150MHz DVB-S: Constellation: QPSK FEC Demodulation: 1/2, 2/3, 3/4, 5/6, 7/8 DVB-S2 : Constellation: QPSK/8PSK/16APSK/32APSK FEC Demodulation: QPSK: 1/2, 2/3, 3/4, 5/6, 3/5, 4/5, 8/9, 9/10 8PSK: 3/5, 2/3, 3/4, 5/6, 8/9, 9/10 16APSK: 2/3, 3/4, 4/5, 5/6, 8/9, 9/10 32APSK: 3/4, 4/5, 5/6, 8/9, 9/10 DVB-S2X : Constellation: QPSK/8PSK/8APSK/16APSK/32APSK FEC Demodulation: QPSK: 1/2, 3/5, 2/3, 3/4, 4/5, 5/6, 8/9, 9/10, 13/45, 9/20, 11/20 8PSK: 3/5, 2/3, 3/4, 5/6, 8/9, 9/10, 23/36, 25/36, 13/18 8APSK: 5/9-L, 26/45-L 16APSK: 2/3, 3/4, 4/5, 5/6, 8/9, 9/10, 1/2-L, 8/15-L, 5/9-L, 26/45, 3/5, 3/5-L, 28/45, 23/36, 2/3-L, 25/36, 13/18, 7/9, 77/90 32APSK: 3/4, 4/5, 5/6, 8/9, 2/3-L, 32/45, 11/15, 7/9, 9/10 Support Diseqc function Multiplexing: Max number of managing PID: 256 Function: PID remapping (automatically/ manually), Accurate PCR adjust, PID pass-through Descrambling: BISS Mode: Mode 1, Mode E; up to 120Mbps, 32 BISS Key

4 frequencies Descrambling Module (DX942A)

Stream in: 4 frequencies input (each RF in interface for 2 frequencies locking), F Type, 16 IP inputs (UDP/RTP) from data port, unicast and multicast, IGMP V2/V3 Stream out: 1 MPTS and 16 SPTS output over UDP and RTP/RTSP, Unicast/Multicast DVB-CI: 2 independent common interface slots Tuner input: DVB-C (J.83 A/C)/J.83B/ DVB-T/DVB-T2/ISDB-T switchable Standard: DVB-C (J.83 A/C); J.83B Input Frequency: 60MHz~890MHz Symbol rate: 1000~9000Ksps Constellation: 16/32/64/128/256 QAM; 64/256 QAM for J.83B Standard: DVB-T/T2 Frequency In: 60MHz~890MHz Bandwidth: 5/6/7/8M bandwidth PLP Index: 0~255 for DVB-T2 Standard: ISDB-T Input Frequency: 60-890MHz Multiplexing: Maximum PID Remapping: 256 output pids Function: PID remapping (automatically or manually), generate PSI/ SI table automatically Descrambling: CAM/CI Quantity: 2 BISS Mode: Mode 1, Mode E; up to 120Mbps, 32 BISS Keys

4 ASI/IP Multiplexing Module (DX504)

Module Specifications: ASI inputs/outputs: 4 ASI bi-direction, BNC 75Ω (manually set input or output) IP inputs/outputs: 2 GE Ports in the module (100/1000Mbps self-adaption) and the DHP200 chassis (data ports or other modules) Re-multiplexing: PID remapping, PCR correction, generate PSI/ SI table automatically Stream In: maximum 4 ASI input, 768 IP inputs over UDP/RTP (256 IP inputs thru GE1, GE2 in the module and the DHP300 chassis respectively) Stream Out: maximum 4 ASI output, 4 MPTS output over UDP/RTP thru GE1, GE2 in the module and thru the DHP300 chassis

5 ASI Multiplexing Module (DX505)

Module Specifications: ASI inputs/outputs: 5 ASI bi-direction, BNC 75Ω (manually set input or output) IP inputs/outputs: the DHP300 chassis (data ports or other modules) Stream in: maximum 5 ASI input, 256 IP inputs over UDP/RTP thru the DHP300 chassis Re-multiplexing: PID remapping, PCR correction, generate PSI/ SI table automatically Stream out: maximum 5 ASI output, 5 MPTS out over UDP/RTP thru the DHP300 chassis

IP Multiplexing Module (DX506)

Module Specifications: IP input: 512 SPTS or MPTS input over UDP, RTP, Unicast and Multicast thru GE1 Ethernet Port (100/1000M) Re-multiplexing: PID remapping, PCR correction, generate PSI/ SI table automatically Stream Out: 512 SPTS output over UDP, RTP, Unicast and Multicast through GE2 Ethernet Port (100/1000M)

ASI/IP Mux-Scrambler Module (DX514)

Module Specifications: ASI inputs/outputs: 4 ASI bi-direction, BNC 75Ω IP inputs/outputs: 2 Ethernet Port (100/1000M) Stream connector input Re-multiplexing: PID remapping, PCR correction, generate PSI/ SI table automatically Stream In: maximum 4 ASI input, 32 IP input per channel over UDP/RTP Stream Out: maximum 4 ASI output, 4 IP (MPTS) output over UDP/RTP/RTSP Scrambling: Maximum simulcrypt CA: 4 Standard: ETR289, ETSI 101 197, ETSI 103 197 Connection: Local/remote connection

ASI TS Switch Module (DX515)

Module Specifications: Input: 3 ASI Output: 2 ASI output (output 1 for one of the switch streams out, output 2 for main input stream) TS packet: 188/204 Switch Mode: Manually/Auto Support monitor program status (bitrate, continuous counting, synchronization status) Support channel mode and program mode switching Support to switch back to the main stream manually or automatically once the main stream recovers

16/32 QAM Modulating Module (DX316/DX332)

Module Specifications: Data input: 512*3 IP input over UDP/RTP from 2 GE Ports (RJ45/SFP) and internal stream connector board, unicast/multicast Data output: 16 or 32 IP output over UDP/RTP/RTSP, unicast/multicast, 2 GE Ports (RJ45/SFP) Trans Rate: Max 840Mbps/GE Port RF output (F type): 16/32 channels of multiplexing, scrambling and modulation. Multiplexing: Maximum PID Remapping: 256 output pids per channel Function: PID remapping (automatically or manually), Accurate PCR adjusting, generate PSI/SI table automatically Scrambling: Maximum simulcrypt CA: 6 Standard: ETR289, ETSI 101 197, ETSI 103 197 Connection: Local/remote connection Modulation: Standard: EN300 429/ITU-T J.83A/B/C MER: ≥ 40 dB RF frequency: 50~960MHz, 1KHz step RF output level: -20~+10dBm (87~117 dB μ V), 0.1dB step for all carriers Symbol Rate: 3600~7200Ksps, 1ksps stepping 5057Ksps (J.83B, 64QAM), 5361Ksps (J.83B, 256QAM) Constellation/Bandwidth: J.83A: 16/32/64/128/256QAM, 8M J.83B/C: 64/256QAM, 6M DX316 Output: 16 non-adjacent carrier outputs within 192M bandwidth DX332 Output: 32 non-adjacent carrier outputs within 384M bandwidth

48 QAM Modulating Module (DX348D)

Module Specifications: Data input: 512*3 IP input over UDP/RTP from 2 GE Ports (RJ45/SFP) and internal stream connector board, unicast/multicast Data output: 48 IP output over UDP/RTP/RTSP, unicast/multicast, 2 GE Ports (RJ45/SFP), GE1 for IP #1-#24, GE2 for IP #25-#48 Trans Rate: Max 840Mbps/GE Port Multiplexing: Maximum PID Remapping: 256 output PIDs per channel Function: PID remapping (automatically or manually), Accurate PCR adjusting, generate PSI/SI table automatically Scrambling: Maximum simulcrypt CA: 6 Standard: ETR289, ETSI 101 197, ETSI 103 197 Connection: Local/remote connection Modulation: Standard: EN300 429/ITU-T J.83A/B/C MER: ≥ 40 dB RF frequency: 50~960MHz, 1KHz step RF output level: -20~+10dBm (87~117 dB μ V), 0.1dB step for all carriers Symbol Rate: 3600~7200Ksps, 1ksps stepping 5057Ksps (J.83B, 64QAM), 5361Ksps (J.83B, 256QAM) Constellation/Bandwidth: J.83A: 16/32/64/128/256QAM, 8M J.83B/C: 64/256QAM, 6M RF output: 2 F type output ports for 48 non-adjacent carriers, 75 Ω Carrier 1~24 out thru RF1, 25~48 thru RF2, each RF supports 384M bandwidth

8/16 ISDB-Tb Modulating Module (DX306I)

Module Specifications: Data input: 512*3 IP input over UDP/RTP from 2 GE Ports (RJ45/SFP) and internal stream connector board, unicast/multicast Data output: 8/16 IP output over UDP/RTP/RTSP, unicast/multicast, 2 GE Ports (RJ45/SFP) Trans Rate: Max 840Mbps/GE Port RF output (F type): 8/16 channels of multiplexing and modulation. Multiplexing: Maximum PID Remapping: 256 input per channel Function: PID remapping (automatically or manually), Accurate PCR adjusting, generate PSI/ SI table automatically Modulation: Standard: ARIB STD-B31 Bandwidth: 6M Constellation: QPSK, 16QAM, 64QAM Guard Interval: 1/32, 1/16, 1/8, 1/4 Transmission Mode: 2K, 4K, 8K Code rate: 1/2, 2/3, 3/4, 5/6, 7/8 MER: $\geq 40\text{dB}$ RF frequency: 50~960MHz, 1KHz step RF output level: -20dBm~+10dBm (87~117dB μ V), 0.1dB stepping RF Output: 8/16 non-adjacent carrier outputs within 192M bandwidth Order Guide: DX306i-8: 8 non-adjacent ISDB-T carrier output + 8 IP MPTS output DX306i-16: 16 non-adjacent ISDB-T carrier output + 16 IP MPTS output

8/16 DVB-T Modulating Module (DX308T)

Module Specifications: Data input: 512*3 IP input over UDP/RTP from 2 GE Ports (RJ45/SFP) and internal stream connector board, unicast/multicast Data output: 8/16 IP output over UDP/RTP/RTSP, unicast/multicast, 2 GE Ports (RJ45/SFP) Trans Rate: Max 840Mbps/GE Port Multiplexing: Channel Number: 8/16 multiplexing channels Maximum PID Remapping: 256 input per channel Function: PID remapping (automatically or manually), Accurate PCR adjusting, generate PSI/ SI table automatically Modulation: Standard: ETSI EN300 744 MER: $\geq 40\text{dB}$ RF Frequency: 50~960MHz, 1KHz step Constellation: QPSK/16QAM/64QAM Bandwidth: 6/7/8 MHz Trans mode: 2K/4K/8K FEC: 1/2, 2/3, 3/4, 5/6, 7/8 RF Output Level: -20~+10dbm (for all carriers), 0.1dB stepping RF Output (F type): 8/16 non-adjacent carrier outputs within 192M bandwidth Order Guide: DX308T-8: 8 non-adjacent DVB-T carrier output + 8 IP MPTS output DX308T-16: 16 non-adjacent DVB-T carrier output + 16 IP MPTS output

8 ATSC Modulating Module (DX308AT)

Module Specifications: Data input: 512*3 IP input over UDP/RTP from 2 GE Ports (RJ45/SFP) and internal stream connector board, unicast/multicast Data output: 8 IP output over UDP/RTP/RTSP, unicast/multicast, 2 GE Ports (RJ45/SFP) Trans Rate: Max 840Mbps/GE Port Multiplexing: Channel Number: 8 multiplexing channels Maximum PID Remapping: 256 input per channel Function: PID remapping (automatically or manually), Accurate PCR adjusting, generate PSI/ SI table automatically Modulation: Standard: ATSC A/53 MER: $\geq 40\text{dB}$ RF Frequency: 50~960MHz, 1KHz step Constellation: 8VSB Bandwidth: 6MHz FEC: RS(208 188)+Trellis RF Output Level: -20~+10dbm (for all carriers), 0.1db stepping RF Output (F Type) : 8 non-adjacent carrier outputs within 192M bandwidth

2 HD-SDI Decoding Module (DX702)

Module Specifications: ASI input/output: 2 ASI bi-directions, BNC 75Ω IP Stream input support IP output: 1 MPTS output over UDP, Unicast/Multicast Decoding: Video/Audio Out: 2 HD/SD SDI output Video Format: MPEG-2, MPEG-4 AVC/H.264 Resolution: 480i,480p,576i,576p,720p@50/59.94/60,1080i@50/59.94/60 Chroma: 4:2:0 Audio Format: MPEG1 Layer2, LC-AAC, HE-AAC, AC3 (2.0/5.1), AC3 Passthrough, Support Dual Audio Out Support CC/Subtitle

4 HDMI Decoding Module (DX714A)

Module Specifications: Input: 8 IP (MPTS/SPTS) input over UDP/RTP thru stream connector, Unicast/Multicast Output: 4 HDMI output Decoding: Video/Audio Out: 4 HDMI output with 1 channel stereo audio embedded in each port Video Format: MPEG-2, MPEG-4 AVC/H.264, HEVC/H.265, AVS, AVS+ Audio Format: MPEG 1 Layer 2, LC-AAC, HE-AAC, AC3 (2.0), DRA Resolution: 480i, 480p, 576i, 576p,1280×720_50p, 1280×720_60p, 1920×1080_50i, 1920×1080_50p 1920×1080_60i, 1920×1080_60p Support manually upscale/downscale resolution

Equipment Specifications:

Base Unit Parameters

Dimension(W×L×H): 440mm×324mm×44mm
Approx weight: 6kg
Environment: 0~45■(work); -20~80■ (Storage)
Power requirements AC 110V± 10%, 50/60Hz, AC 220 ± 10%, 50/60Hz

About Us

FMUSER is a company specializing in broadcasting and television transmission systems and integrating hotel IPTV systems. The company was established in 2009 and has served over 180 countries worldwide. More than 20,000 television stations, radio stations, and hotel clients from different.

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